

COMP122 Portfolio Projects

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1 Introduction

The goal of the portfolio project is to build a Java application that demonstrates the programming skills and concepts you have learned in this course. By the end of the project, you will have a substantial piece of work that showcases your ability to design, implement, and test a Java program. It can also serve as part of your portfolio when applying for internships or future jobs.

We encourage you to choose a project that interests you and allows you to be creative. The more engaged you are with your idea, the more enjoyable the project will be.

- You may begin working on the portfolio project only after you have passed Mastery Tests 1–4.
- You may complete the project either individually or with one teammate.

If you would like to create your own project, please discuss your project idea with the teaching staff before you begin. Prepare a brief description similar to the example project descriptions (in Section 2) so we can ensure that the project is within the expected scope (i.e. neither too simple nor too complex). Your project must incorporate inheritance, recursion, and multiple data structures.

If you are unable to come up with your own idea, you may choose one of the example portfolio projects mentioned in Section 2.

2 Example Projects

2.1 Volcano Monitoring & Eruption Prediction System

New Zealand is home to several active volcanoes, including Ruapehu, Ngāuruhoe, Taupō, Rotorua, and Rangitoto. These volcanoes differ significantly in their geological characteristics, eruption styles, and associated hazards. Scientists continuously monitor volcanic activity using seismic sensors, gas measurements, ground deformation, and historical eruption records to determine the likelihood of future eruptions and issue public warnings.

This project will develop an object-oriented Volcano Monitoring & Eruption Prediction System. The system will model different types of volcanoes, process monitoring data, maintain eruption histories, estimate eruption risk and manage emergency notifications.

System Overview

This project will store information about five volcanoes: Ruapehu, Ngāuruhoe, Taupō, Rotorua, and Rangitoto. Ruapehu and Ngāuruhoe are stratovolcanoes, which are steep, cone-shaped volcanoes that typically produce explosive eruptions due to trapped gases. Taupō and Rotorua are caldera volcanoes, which are large, lake-filled volcanic systems associated with significant geothermal activity. Rangitoto is a shield volcano with broad, gentle slopes that primarily produces lava eruptions.

For each volcano, the system must store its name, current GeoNet alert level (0–5), historical eruption records, and the last 10 seismic intensity readings. The system must support calculating the volcano's current eruption risk, adding new eruption records and seismic readings, and displaying the volcano's current information.

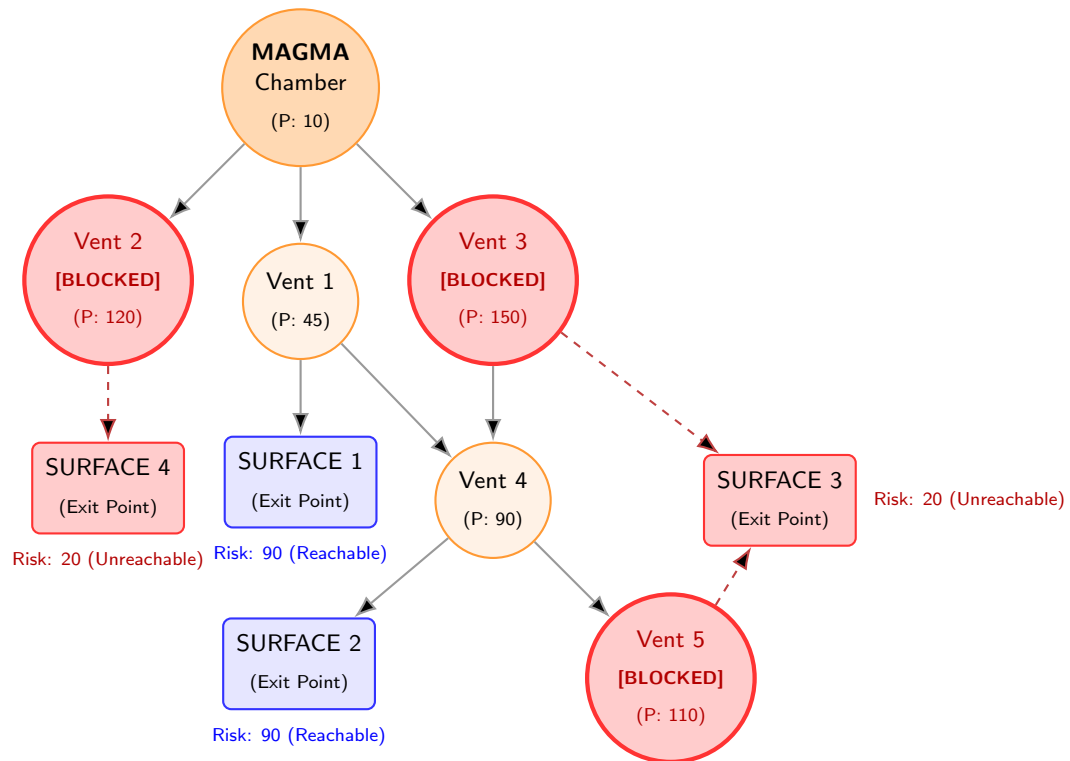
The current eruption risk score ranges from 0 (safe) to 100 (imminent eruption) and is calculated differently depending on the volcano type:

- Stratovolcano risk depends on the current gas pressure (≥ 0) and seismic intensity readings:

$$\text{risk} = (\text{gasPressure} \times 0.5) + (\text{avgSeismicReading} \times 0.5)$$

where *avgSeismicReading* is the average of the last 10 seismic intensity readings. Each seismic intensity reading is ≥ 0 .

- Caldera volcano risk is determined by whether magma can travel through a network of underground vents and successfully reach the surface.



A volcano's magma chamber is connected to the surface through a network of underground vents. Each vent has:

- a pressure value; and
- a collection of connected vents that magma can travel to.

The number of connected vents can vary. A vent may have no connected vents (a surface vent) or may connect to multiple other vents.

Magma starts at the magma chamber (the root vent) and can move through the vent network by travelling from one vent to any of its connected vents.

However, a vent is considered blocked if its pressure is greater than 100. If magma reaches a blocked vent, that path cannot be used and the algorithm must backtrack to explore alternative routes.

A vent with no connected vents represents a possible exit to the surface. Magma successfully reaches the surface if it reaches an unblocked surface vent.

The system should recursively explore all possible paths through the vent network to determine whether magma can successfully reach the surface.

The risk is calculated as:

$$risk = \begin{cases} 90, & \text{if magma can reach the surface} \\ 20, & \text{if magma cannot reach the surface} \end{cases}$$

- Shield volcano risk is primarily determined by lava flow rate (≥ 0) and ground inflation (between 0 and 100):

$$risk = (lavaFlowRate \times 0.4) + (groundInflation \times 0.6)$$

The NZ GeoNet alert level is determined from the calculated risk score:

- Level 0 (No volcanic unrest): $risk < 15$
- Level 1 (Minor volcanic unrest): $risk < 35$
- Level 2 (Moderate to heightened volcanic unrest): $risk < 60$
- Level 3 (Minor volcanic eruption): $risk < 75$
- Level 4 (Moderate volcanic eruption): $risk < 90$
- Level 5 (Major volcanic eruption): $risk \geq 90$

When the NZ GeoNet alert level is 3 or higher, the system broadcasts a warning for that volcano. However, sensor errors may occasionally produce false alarms. Therefore, the system must support undoing the most recent alert and displaying all currently active alerts, starting from the most recent alert.

The volcanoes can be ordered from most to least dangerous based on their eruption risk score.

2.2 Dragon's Hoard Manager

You are a dragon. A very rich, very possessive dragon. Manage your treasure hoard and hired security forces to ensure that your treasures remain safe from sneaky adventurers!

System Overview

The two types of treasure currently stored in your hoard are gemstones and relics. For each treasure item, you store its name, origin, and a list of its 10 most recent appraisal values. However, you also have your own formula for determining how valuable each treasure is to you.

- A gemstone's value is calculated based on its weight in carats (> 0), clarity score (1 to 10), and average appraisal value.

$$value = (weight \times clarityScore \times 50) + avgAppraisal$$

where *avgAppraisal* is the average of the last 10 appraisal values. Each appraisal value is ≥ 10 .

- Relics are hidden deep inside a labyrinth of interconnected tunnel chambers.

The tunnel system is structured as a network, where each tunnel chamber contains:

- a danger level representing the risks faced by intruders, such as traps, lava pits, or fragile ceilings; and
- a collection of connected tunnel chambers that an intruder can move to.

Each chamber can have a different number of connected chambers. Some chambers may have no connections, representing the possible locations of hidden relics, while others may connect to multiple chambers.

An intruder starts at the entrance chamber and can move through the tunnel network by travelling from one chamber to any of its connected chambers.

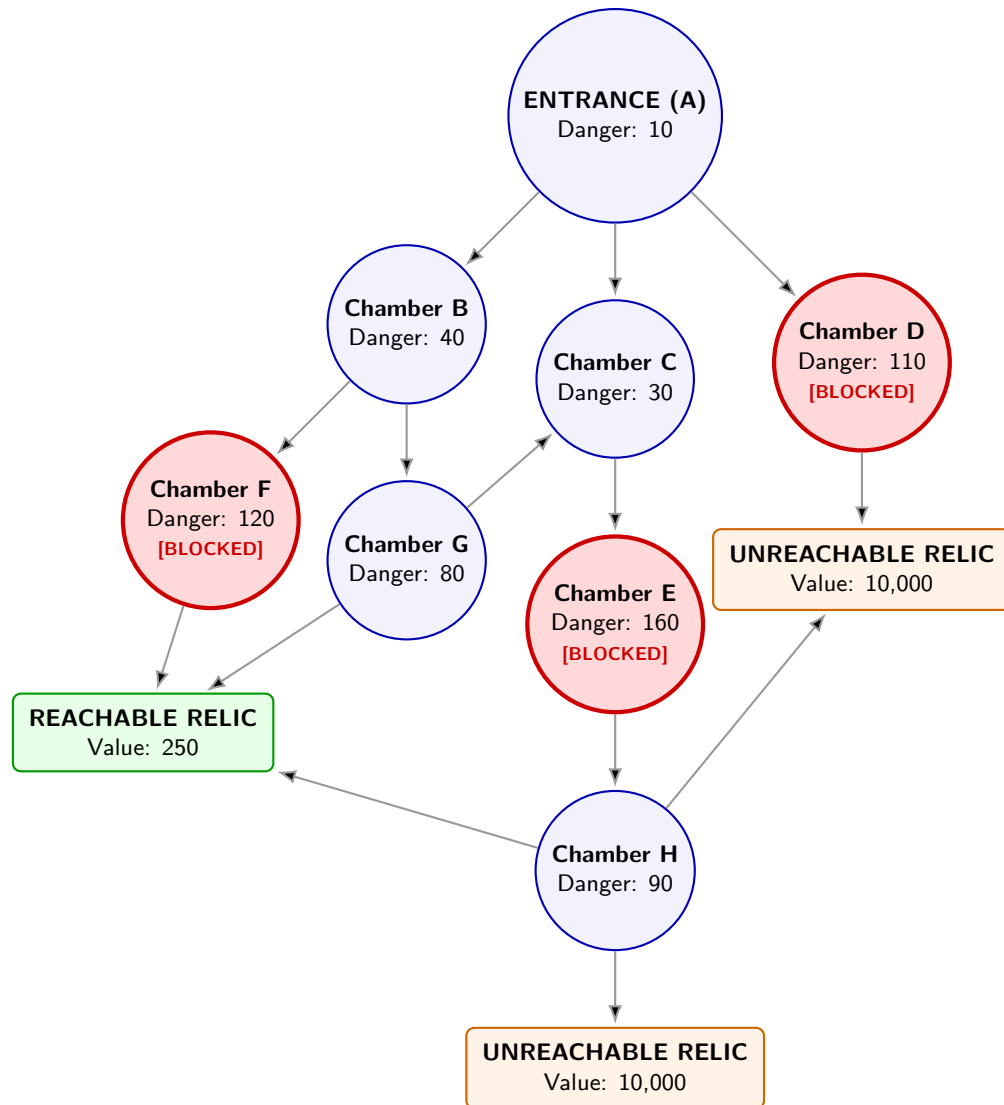
A chamber is considered blocked if its danger level is greater than 100. If an intruder reaches a blocked chamber, that path cannot be used, and the algorithm must backtrack to explore alternative routes.

The system should recursively explore all possible paths through the tunnel network to determine whether an intruder can successfully reach a relic hidden at the end of the tunnel network.

The value of a relic is calculated as:

$$value = \begin{cases} 250, & \text{if an adventurer can reach the relic} \\ 10,000, & \text{if an adventurer cannot reach the relic} \end{cases}$$

The treasure items can be sorted based on their value to the dragon, from highest to lowest.



The security threat level (*threatLevel*) of a treasure item, representing the potential danger if it is stolen by invading adventurers, is determined based on its value:

- 0 (no action) if *value* < 100
- 1 (log only) if *value* < 500
- 2 (watch status) if *value* < 1,500
- 3 (arm passive traps) if *value* < 5,000
- 4 (arm lethal traps) if *value* < 10,000
- 5 (full lair lockout) if *value* ≥ 10,000

When a treasure item's *threatLevel* is ≥ 3 , the dragon must activate traps to protect it.

To further protect the lair, the dragon has hired several mercenary squads. Each squad has a name, a number of guards, and a daily contract rate. The value of a mercenary squad is calculated as:

$$value = guardCount \times dailyContractRate \times 30$$

The dragon does not care about protecting mercenary squads because they are considered replaceable (*threatLevel* = 0), so no traps are required for them.

The dragon maintains a record of deployed mercenary squads protecting its precious hoard. When a squad is assigned to guard the lair, it is added to the top of the deployment order. However, dragons are famously protective of their gold — and not very fond of paying unnecessary wages. When the dragon decides that it has hired too many expensive heroes standing around doing nothing, temporary reinforcement squads are dismissed in reverse order, with the most recently deployed squad sent home first.

3 Software Testing

You are expected to thoroughly test the functionality of your application using a variety of inputs. For the example projects, we have provided sample inputs and their expected outputs, which you can view in Aoroa. These examples are provided as a reference, but you should design additional test cases to ensure that your application works correctly under different scenarios.

4 Marking Criteria

The portfolio project is worth 8% of your final grade and is assessed using the following criteria:

- Code design (1 mark)
- Code implementation (1 mark)
- Code quality (structure, comments, and exception handling) (1 mark)
- Software testing (1 mark)
- Ability to explain your design decisions (1 mark)*
- Ability to explain your implementation (1 mark)*
- Ability to correctly modify your code during an in-person lab assessment (2 marks)*

Half of your mark is based on your submitted code, and the other half is based on an in-person assessment (indicated with *). During the lab, you will explain your design and implementation decisions, demonstrate your understanding by making changes to your code when requested by the teaching staff, and explain the time and space complexity of selected methods.

The code implementation and in-person assessment are marked independently. Therefore, even if your implementation is incomplete or does not fully meet the specification,

you can still achieve full marks in the in-person assessment by demonstrating a thorough understanding of your solution, explaining the challenges you encountered, describing how you planned to address them, and explaining why your implementation did not behave as intended.

If you are working with a teammate, you must clearly identify each person's contributions. The in-person assessment is completed and marked individually. Each team member will be assessed only on the parts they designed and implemented.

Please ensure that your application:

- follows good object-oriented design and programming principles (e.g., modular design, one responsibility per method, meaningful class/method/variable names, and no duplicated code)
- uses appropriate data structures
- includes clear and complete Javadoc comments
- is thoroughly tested, including edge cases
- handles exceptions gracefully (e.g., invalid user input)

5 Submission and Deadline

The portfolio project is due by **5:00 pm on Friday, 16 October 2026**.

Submit a ZIP file to Aoroa containing:

- the source code for all Java classes (in the correct folder structure)
- the code and files used to test your application
- the test cases used to evaluate your application, including the input, the actual output generated by your application, the expected output, and a brief explanation of whether your application produced the expected results.
- instructions explaining how to compile, run, and test your application

You may complete the in-person assessment **only after** submitting your ZIP file to Aoroa. The assessment will take place during your scheduled lab session between **Monday, 12 October 2026** and **Friday, 16 October 2026**.

We will aim to assess you as soon as possible after you submit your portfolio. However, because of the large number of students, this may not always be possible. To maximise your chance of completing the in-person assessment before the deadline, we strongly recommend submitting your ZIP file by **9:00 am on Monday, 12 October 2026**. Earlier submissions are encouraged.

6 Late Submission

To allow you to focus on your final exam preparation, there will be **no in-person assessments** after the deadline.

Late submissions will incur a penalty of **5% of the available marks per day**.

7 Academic Integrity

Students are expected to be aware of, and act in accordance with, the University's Academic Integrity Policy. All submitted work for assessment must be your own.

Academic Misconduct, such as plagiarism or cheating, is a breach of Academic Integrity and is taken very seriously by the University. Types of misconduct include plagiarism, copying, unauthorised collaboration, impersonation, and assisting someone else's misconduct. A more extensive list of the types of academic misconduct and associated processes and penalties is available in the University's Student Academic Misconduct Procedures.

Use of generative software such as ChatGPT is not allowed for this assessment. You are expected to develop the solution independently by thinking through every step of the design and implementation process yourself. This includes determining the required classes, data fields, and methods, designing an efficient and appropriate implementation for each task, and deciding how to evaluate and test your code to ensure that it is correct and robust. The purpose of this assessment is to assess your ability to analyse a problem, design a solution, implement it using the programming concepts covered in this course, and verify that your solution works as intended.

It is your responsibility to be aware of and use acceptable academic practices when completing your assessments. To access the information in the Academic Integrity Policy and learn more, please visit the University's Academic Integrity website at www.otago.ac.nz/study/academicintegrity or ask at the Student Learning Centre or Library. If you have any questions, ask your lecturer.

Links to the policies, for your reference:

- Academic Integrity Policy
<http://www.otago.ac.nz/administration/policies/otago116838.html>
- Student Academic Misconduct Procedures
<http://www.otago.ac.nz/administration/policies/otago116850.html>